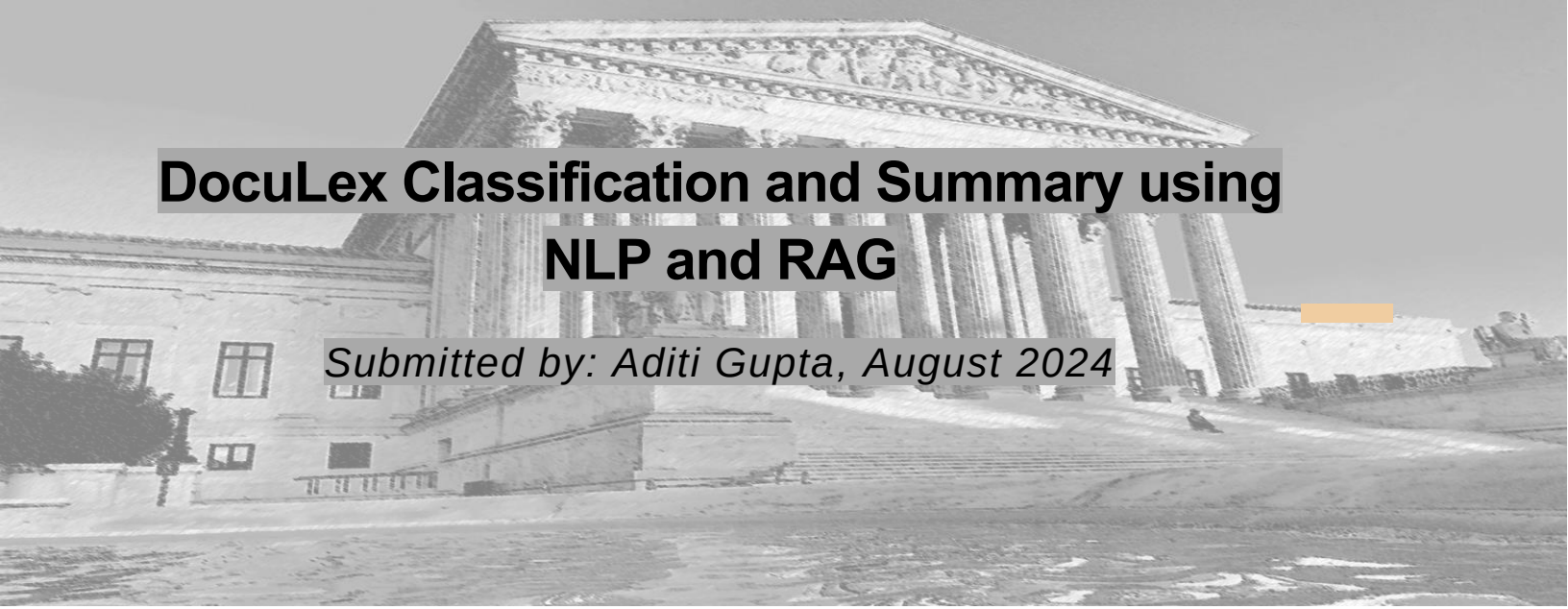




DocuLex Classification and Summary using NLP and RAG

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INTRODUCTION

Legal professionals are inundated with a vast array of complex documents, including contracts, case law, judicial opinions, and oral arguments. The manual process of analyzing and summarizing these documents is not only time-consuming but also prone to errors, leading to inconsistencies and inefficiencies in legal research.

This project leverages the SCOTUS dataset, which contains a wealth of legal documents from the U.S. Supreme Court, to apply Natural Language Processing (NLP) techniques for automating and enhancing the analysis of these documents. By developing a system capable of efficiently extracting key information, summarizing content, and classifying documents based on their content, such as contract types or case law topics, we aim to save time and increase the accuracy and consistency of legal document analysis.

DATASET

This project utilizes the SCOTUS dataset, a comprehensive collection of legal documents from the U.S. Supreme Court. The dataset includes various attributes critical to the analysis, such as:

Column Name	Description
id	Unique identifier for each case.
name	The full name of the case.
name_abbreviation	An abbreviated version of the case name.
decision_date	The date when the decision was made, formatted in ISO format (YYYY-MM-DD) or other date formats like MM/DD/YYYY.
first_page	The first page number of the case document.
last_page	The last page number of the case document.
cites_to	References to other legal cases or documents.
last_updated	The date when the record was last updated.
file_name	The name of the file containing the case information.
first_page_order	Order of the first page in the document.
last_page_order	Order of the last page in the document.
court.name_abbreviation	Abbreviation of the court's name.
court.id	Unique identifier for the court.
court.name	Full name of the court.
jurisdiction.id	Unique identifier for the jurisdiction.
jurisdiction.name_long	Long form of the jurisdiction's name.
jurisdiction.name	Short form of the jurisdiction's name.
analysis.cardinality	Cardinality of the analysis, which may refer to the number of unique elements or instances.
analysis.char_count	Number of characters in the analysis text.
analysis.ocr_confidence	Confidence level of the Optical Character Recognition (OCR) process.
analysis.pagerank.raw	Raw PageRank score from the analysis.
analysis.pagerank.percentile	Percentile ranking of the PageRank score.

analysis.sha256	SHA-256 hash of the document for integrity verification.
analysis.simhash	SimHash value for document similarity comparison.
analysis.word_count	Number of words in the analysis text.
provenance.date_added	Date when the record was added to the dataset.
provenance.source	Source of the data.
provenance.batch	Batch or group in which the data was processed.
casebody.judges	Names of the judges involved in the case.
casebody.parties	Names of the parties involved in the case.
casebody.opinions	Opinions or judgments related to the case.
casebody.attorneys	Names of the attorneys involved in the case.
casebody.head_matter	Head matter of the case, including information such as the case name, court, and decision date, formatted as month day, year.
citations_type	Type of citations used in the case.
citations_cite	Specific citations referenced in the case.
volume_number	Volume number of the case report.
publisher	Publisher of the case report.
publication_year	Year the case was published.
start_year	Year when the case or related publication started.
end_year	Year when the case or related publication ended.
jurisdictions	Jurisdictions relevant to the case.
harvard_hollis_id	Identifier used by the Harvard Hollis catalog.
spine_start_year	Starting year for the spine of the case report.
spine_end_year	Ending year for the spine of the case report.
publication_city	City where the case was published.
redacted	Indicates if the document has been redacted.

PROBLEM STATEMENT

The project aims to address the question: *Can key information be automatically extracted from legal documents, and can these documents be accurately classified and summarized using NLP techniques to aid legal professionals in their research and decision-making processes?*

The complexity and variability within the SCOTUS dataset necessitate advanced NLP techniques to automate the extraction, classification, and summarization of legal documents, thereby streamlining the workflow for legal professionals.

DATA WRANGLING

Data wrangling involved several crucial steps:

- 1. Data Import and Conversion:**

Imported 180-181 JSON files and converted them into a DataFrame. Given the complexity of nested JSON structures, normalization of columns with complicated structures was performed separately. Load times were calculated to plan the data import process efficiently.

2. Data Exploration and Cleansing:

Conducted an exploratory analysis using methods such as `head()`, `info()`, `shape`, and `describe`. The dataset was cleansed by counting null values, identifying unique values per column, and removing rows with irrelevant NaN values. Columns with a significant number of NaN values were also dropped.

3. Feature Engineering:

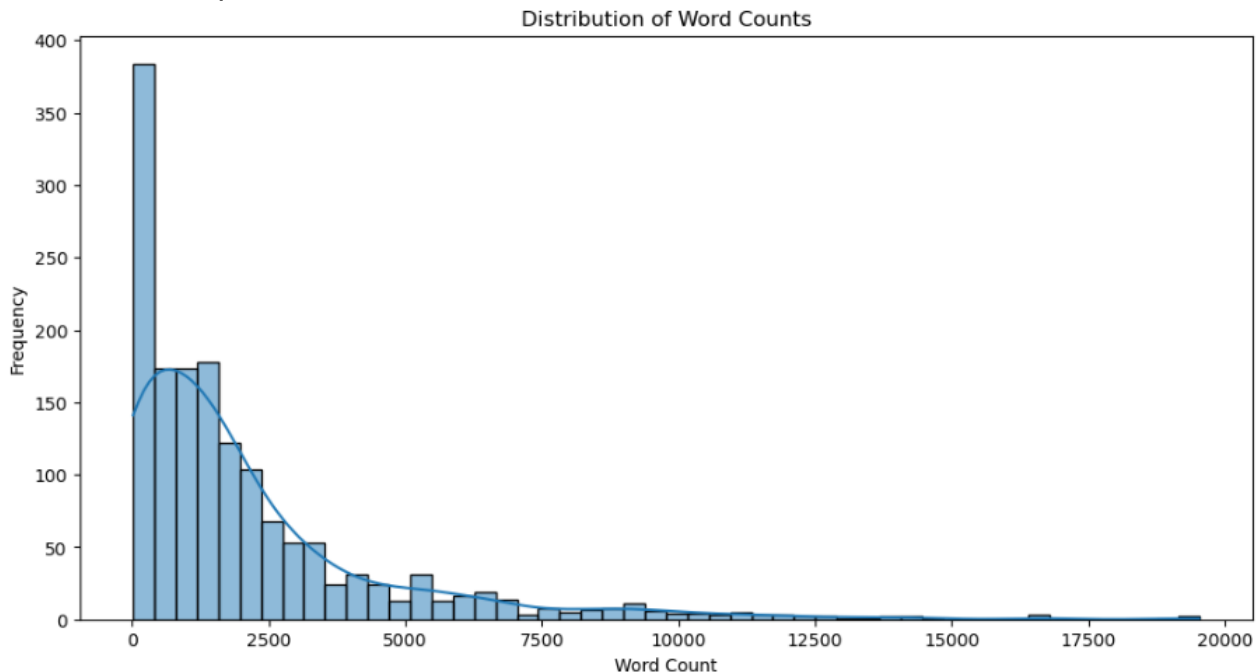
The `casebody_judge` column was enriched by extracting the last author from `casebody.opinions` and merging it with `casebody.judges`. Additionally, the `case_length` was calculated by converting `first_page` and `last_page` to numeric values.

EXPLORATORY DATA ANALYSIS

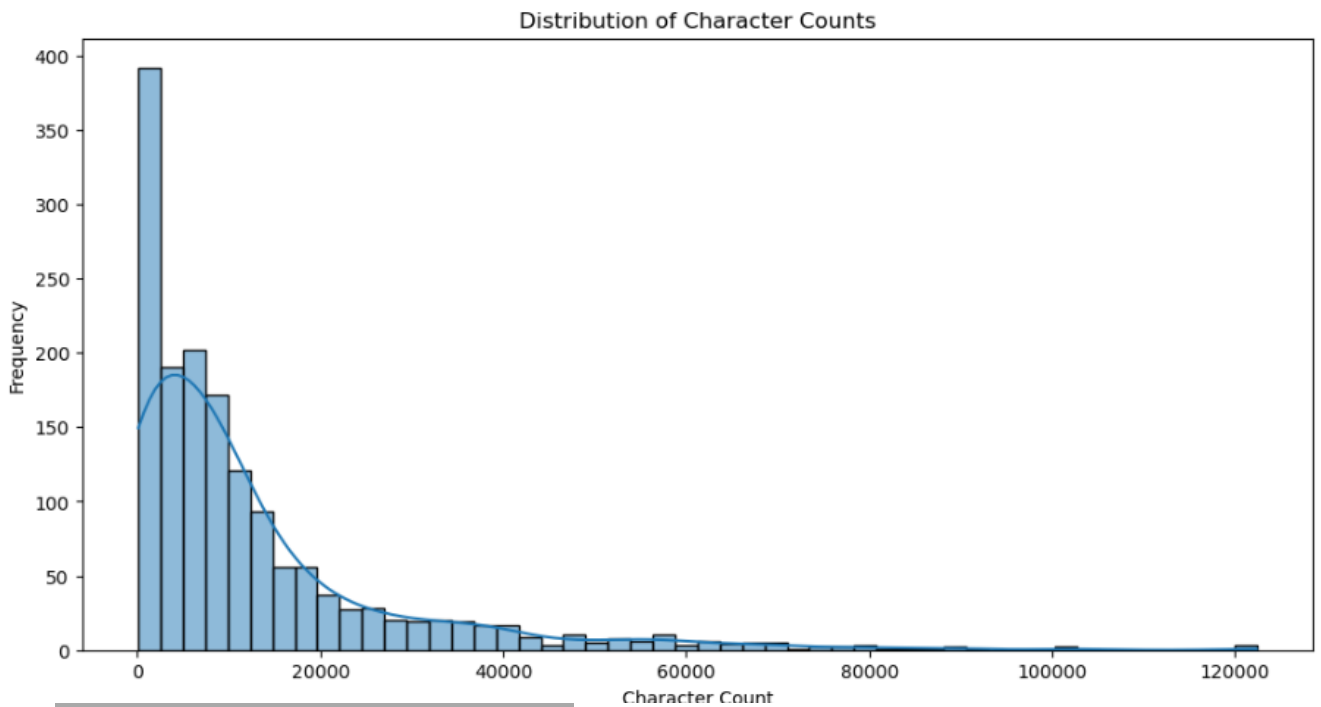
EDA was essential in uncovering insights and patterns within the SCOTUS dataset. The following analyses provided a deeper understanding of the data:

1. Univariate Analysis: Analyzed the distributions of character counts, case length, and OCR confidence levels across the dataset.

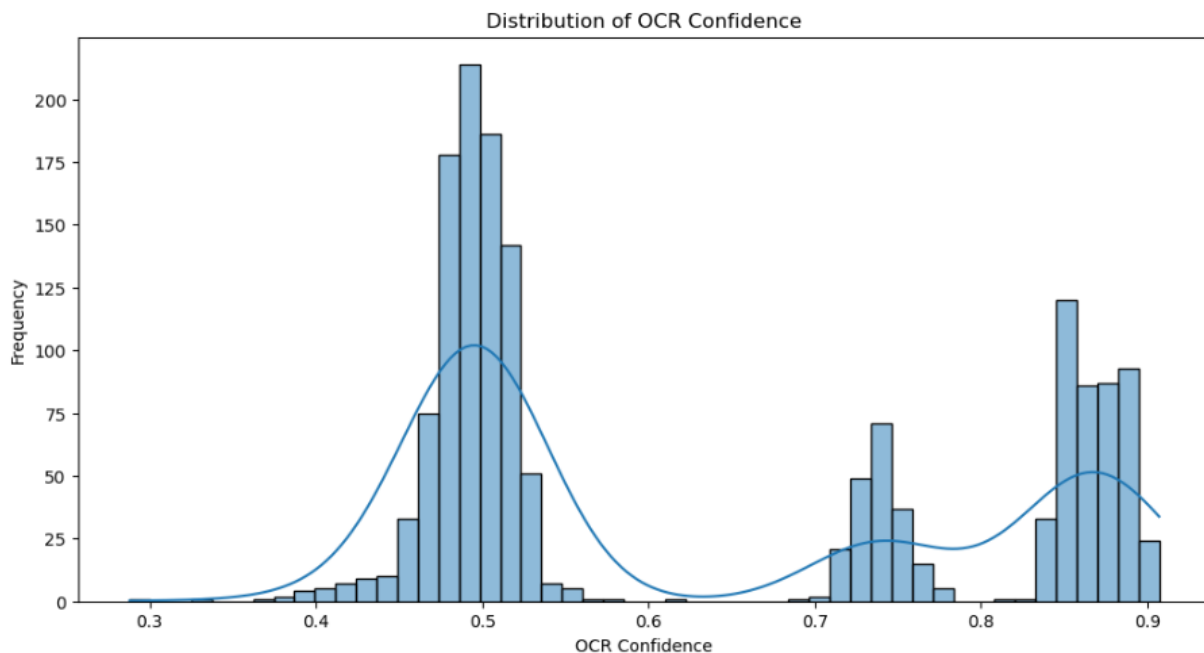
- a) **# Distribution of word counts:** The distribution of word counts shows a significant peak at 0, indicating many documents contain very few words, likely metadata or short summaries. The long tail extending up to 19,000 words suggests the presence of a few documents that are exceptionally long, potentially detailed opinions or case summaries.



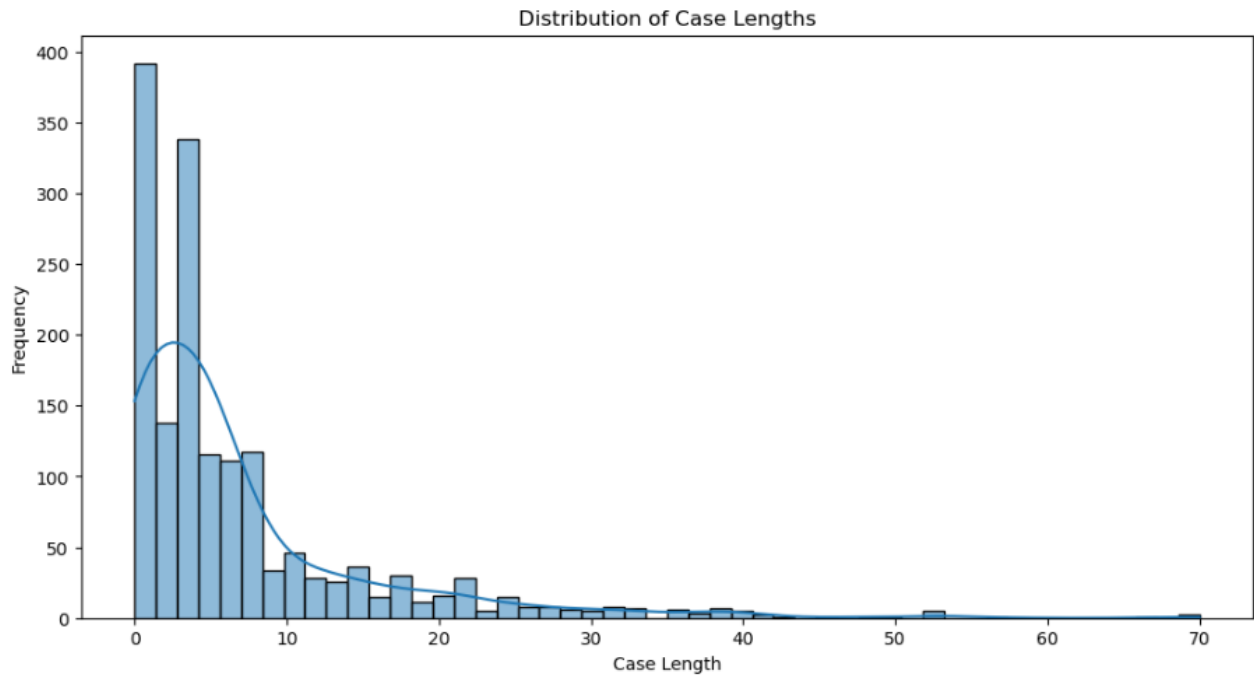
- b) **# Distribution of character counts** Character count distribution mirrors the word count, with a peak at 0, reflecting the high occurrence of short texts. The tail reaching up to 120,000 characters indicates that while most documents are brief, some are significantly lengthy, likely containing detailed legal opinions or judgments.



c) # Distribution of OCR confidence OCR confidence is distributed across multiple peaks, particularly around 0.5, 0.7-0.8, and 0.8-0.9. This suggests that the OCR process had varying levels of accuracy, with some documents being more accurately digitized than others. The peaks indicate common levels of OCR accuracy across different document types.



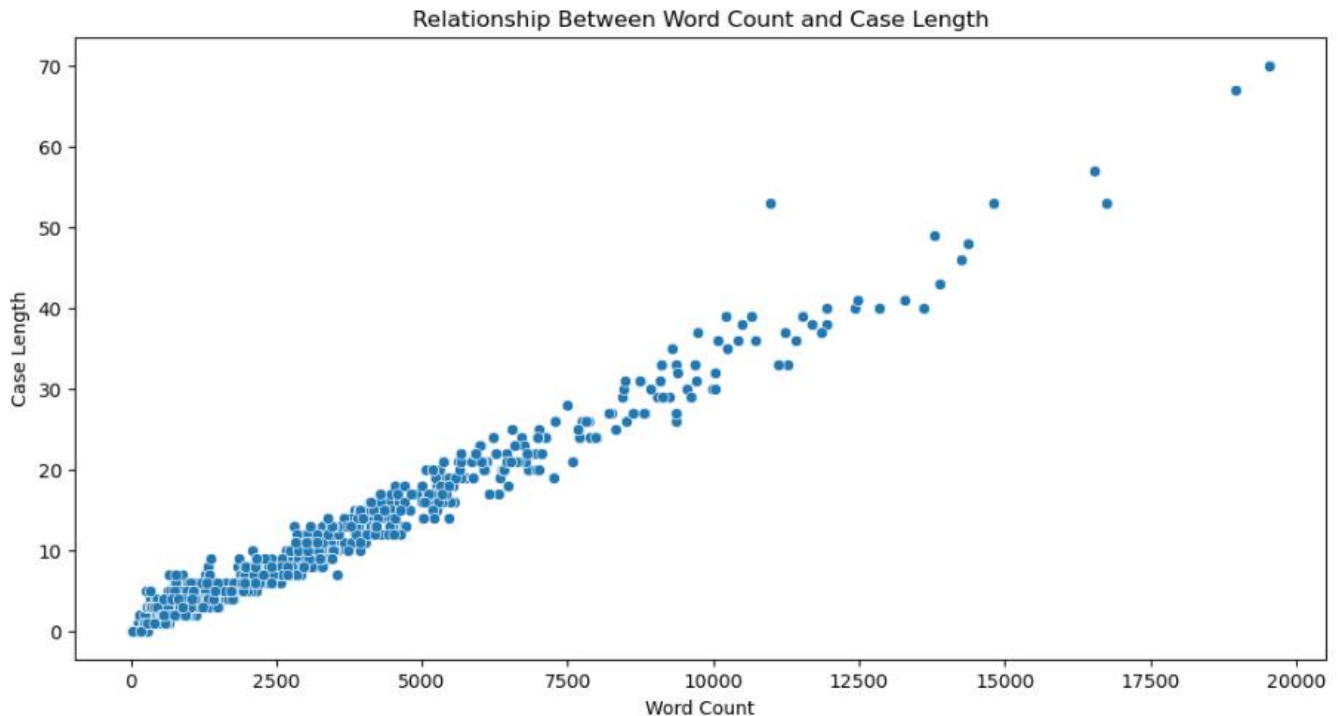
d) # Distribution of case lengths The case length distribution peaks at 0, indicating that many cases are very short. The gradual decline and long tail up to 70 pages suggest that while most cases are brief, there are outliers that are significantly longer, likely complex cases or those with extensive documentation.



2. Bivariate Analysis

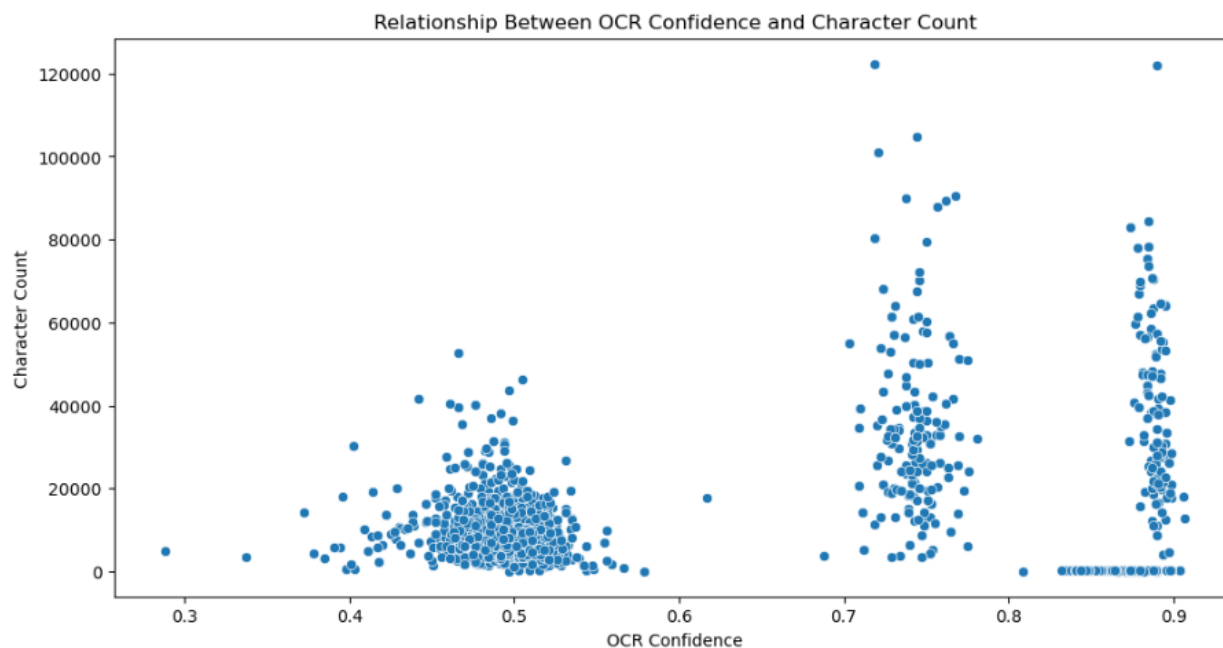
Explored relationships between key variables.

- a) **Relationship Between Word Count and Case Length**: The positive correlation between word count and case length is evident in this plot. As the number of words in a document increases, so does the case length, suggesting that longer legal documents naturally span more pages, likely due to the complexity and detail involved.



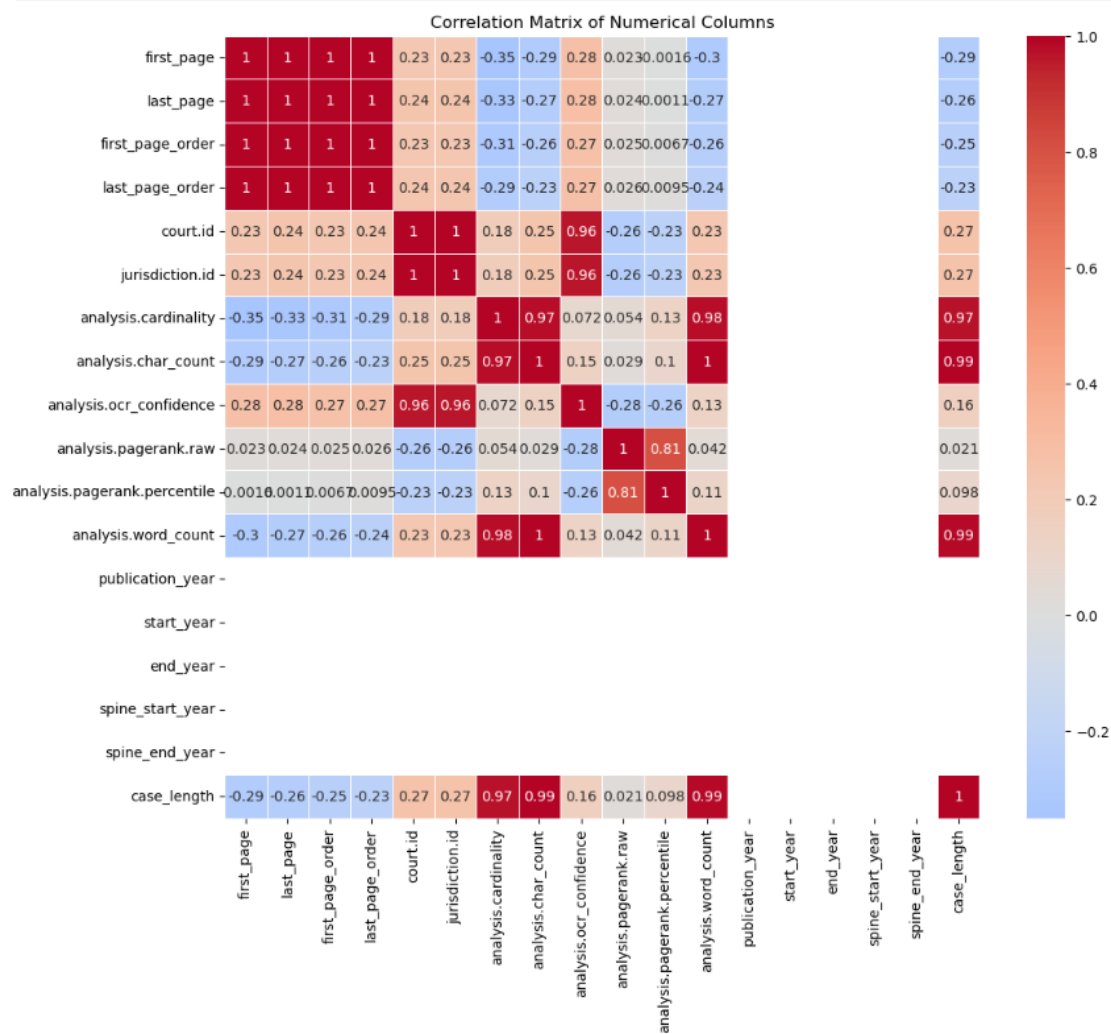
- b) **Relationship Between OCR Confidence and Character Count**: The scatter plot revealed three distinct clusters that correlate OCR confidence with character count. Documents with lower OCR confidence tend to be shorter, whereas those

with higher confidence generally have a broader range of character counts. The high confidence clusters with short documents suggest well-documented cases, while outliers with high character counts and varying OCR confidence may indicate anomalies or particularly challenging documents for OCR.



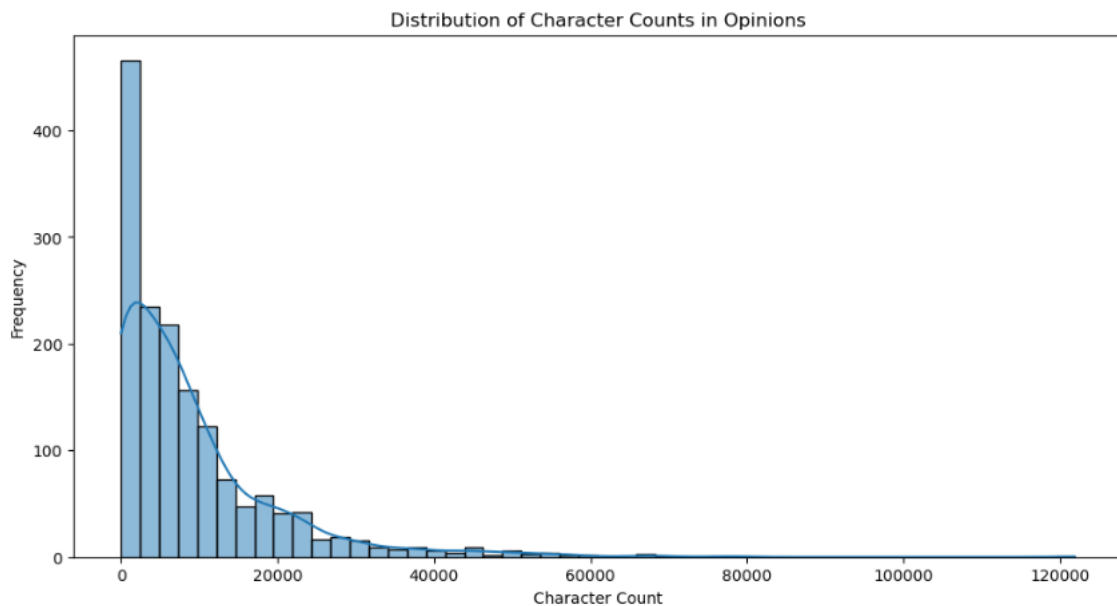
3. Correlation Analysis

- e) The correlation matrix provided insights into the relationships between different numerical variables in the dataset. For example, a strong correlation between word count and case length confirms the finding from the bivariate analysis, indicating that these two features are closely related.
- f) Correlation Matrix for Numerical Columns



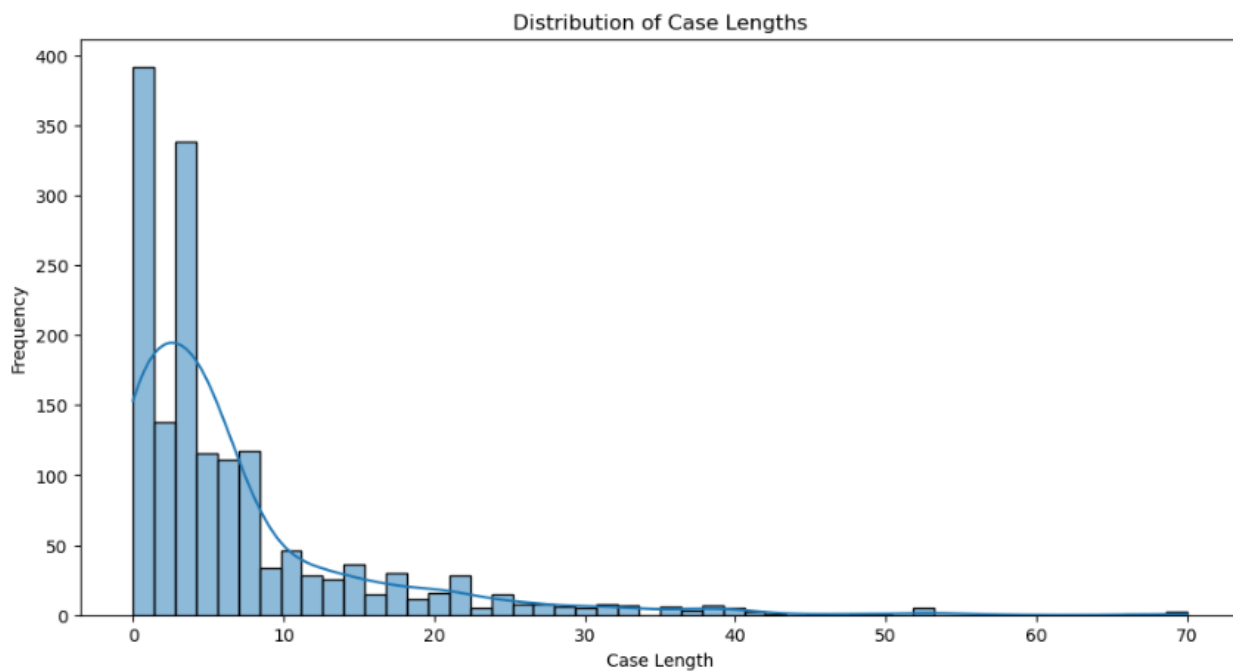
4. Temporal Analysis of Case Trends Over Time: Tracks the number of cases decided each year to identify trends over time.

- g) **Number of Cases Decided Each Year** The temporal analysis tracks the number of cases decided each year, showing fluctuations in case volumes over time. Peaks in certain years may reflect periods of heightened legal activity or significant judicial events, while declines might indicate years with fewer cases or reporting inconsistencies.



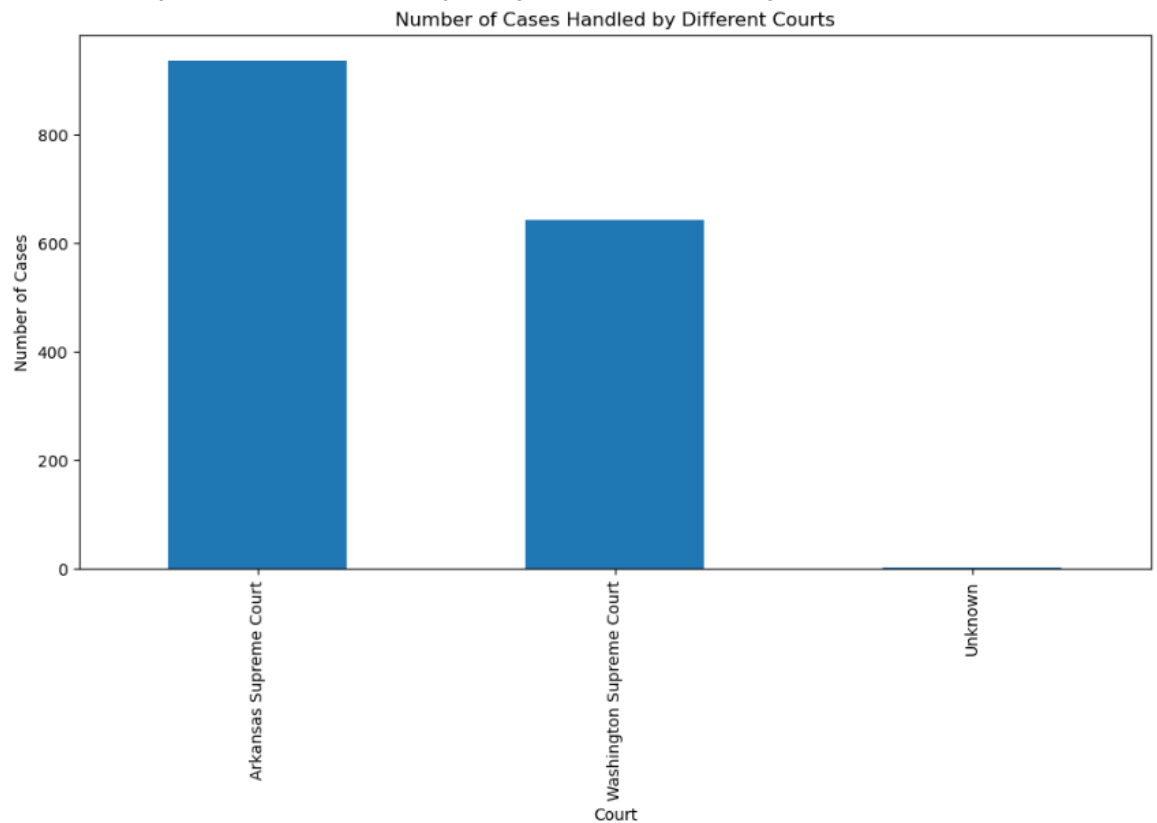
6. Case and Court Analysis

- a) **Case Lengths and Citation Counts:** This analysis investigates the relationship between the length of cases and the number of citations they contain. Longer cases often include more citations, reflecting the extensive legal precedents considered in complex cases.



- b) **Court Analysis:** This analysis assesses the number of cases handled by different courts and jurisdictions. It reveals the distribution of judicial activity, with some courts handling a higher volume of cases, likely reflecting their

jurisdictional importance or the complexity of the cases they oversee.

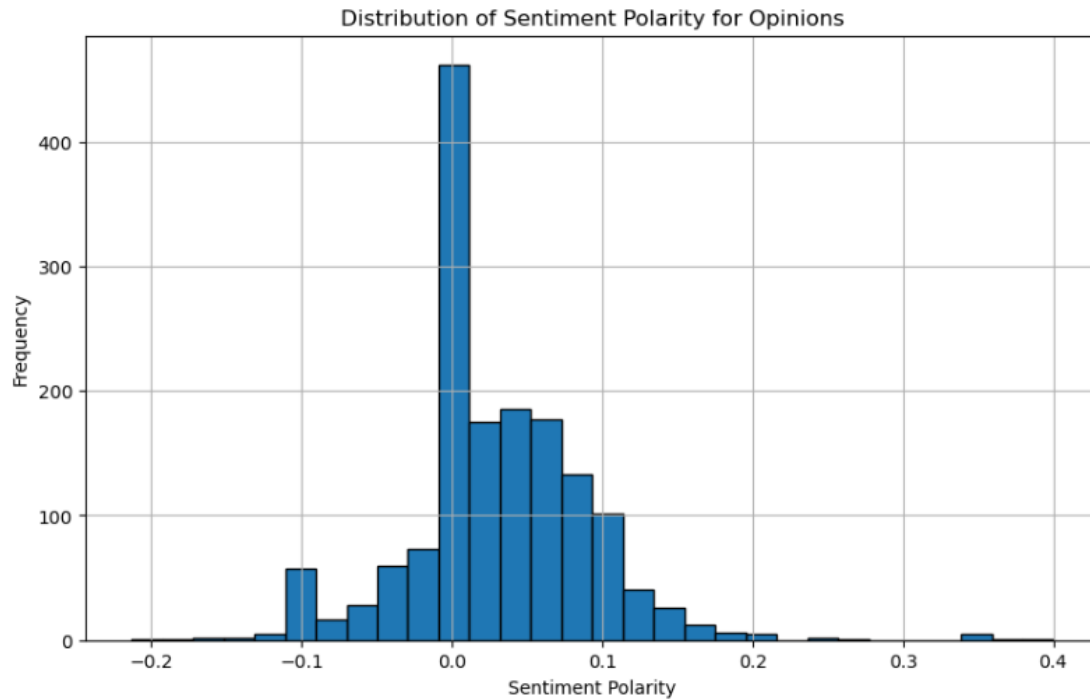


7. **Analysis of Judges and Their Opinions:** Examines the frequency of judges and the sentiment of their written opinions.

- a) **Frequency of Judges** The frequency distribution of judges involved in the cases shows which judges are most active in the dataset. This can provide insights into the judicial workload and the influence of certain judges in legal decision-making.

```
Smith, J.          128
Humphreys, J.     114
Butler, J.        76
McHaney, J.       63
...
Meuaffy, J.       1
Mr. Justice Hanly 1
Macuxn           1
Mehafey, J.      1
McHaNEy, J.      1
Name: casebody_judges, Length: 151, dtype: int64
```

- b) **Sentiment Analysis of Opinions** The sentiment analysis examines the tone of the judicial opinions, providing insights into whether the opinions tend to be more positive, neutral, or negative. This can reflect the judicial approach and the nature of the cases being decided.



PREPROCESSING AND TRAINING

The preprocessing and training phase involved:

- **Feature Selection:**
Removed unnecessary columns with all NaN values or irrelevant data for this project, such as `docket_number`, `casebody.corrections`, `title`, etc.
- **Data Transformation:**
Encoded categorical variables, such as `court.name_abbreviation`, extraction of sub lists from `opinions` column.
- **Data Splitting:**
The data was split into training (80%) and testing (20%) sets to ensure a robust model evaluation.

MODEL GENERATION AND EVALUATION

Three different models were developed and evaluated to address the problem statement:

NER Model:

Focused on identifying the decision date from the `casebody.head_matter` field. Despite the challenge of multiple dates within the text, the model successfully extracted the most probable date, demonstrating strong performance with negligible overfitting.

Model Performance on Training Data:
Accuracy: 99.76%
Precision: 100.00%
Recall: 99.76%
F1 Score: 99.88%

Model Performance on Test Data:
Accuracy: 99.37%
Precision: 100.00%
Recall: 99.37%
F1 Score: 99.68%

Overfitting Check:
Difference in Accuracy: 0.0039
Difference in Precision: 0.0000
Difference in Recall: 0.0039
Difference in F1 Score: 0.0020

Text Summarization (BART):

Generated case summaries using the case name and evaluated them against the case name abbreviation as a baseline. The model showed promising results with high recall but lower precision and F1 scores, indicating potential areas for improvement in summary quality.

Aggregated ROUGE Scores:
Rouge1 - Precision: 0.1279, Recall: 0.9099, Fmeasure: 0.2200
Rouge2 - Precision: 0.0757, Recall: 0.5974, Fmeasure: 0.1309
RougeL - Precision: 0.1279, Recall: 0.9099, Fmeasure: 0.2200
Original Text 84: West Twelfth Street Improvement District No. 30 v. Kinstley
Name Abbreviation 84: West Twelfth Street Improvement District No. 30 v. Kinstley
Generated Summary 84: West Twelfth Street Improvement District No. 30 v. Kinstley was a case in which the court ruled in favor of the Improvement District. The case was heard in the U.S. District Court for the Eastern District of New York.
ROUGE Scores 84: {'rouge1': Score(precision=0.225, recall=1.0, fmeasure=0.36734693877551017), 'rouge2': Score(precision=0.20512820512820512, recall=1.0, fmeasure=0.3404255319148936), 'rougeL': Score(precision=0.225, recall=1.0, fmeasure=0.36734693877551017)}

Original Text 54: Producers' Sand & Gravel Company, Inc., v. Patterson
Name Abbreviation 54: Producers' Sand & Gravel Co. v. Patterson
Generated Summary 54: Producers' Sand & Gravel Company, Inc., v. Patterson, is a case involving sand and gravel mining in the United States. The case was decided in favor of the producers, and the decision was appealed.
ROUGE Scores 54: {'rouge1': Score(precision=0.15151515151515152, recall=0.8333333333333334, fmeasure=0.25641025641025644), 'rouge2': Score(precision=0.09375, recall=0.6, fmeasure=0.16216216216216214), 'rougeL': Score(precision=0.15151515151515152, recall=0.8333333333333334, fmeasure=0.25641025641025644)}

Original Text 71: Lefevers v. Dierks Lumber & Coal, Company
Name Abbreviation 71: Lefevers v. Dierks Lumber & Coal, Co.
Generated Summary 71: Lefevers v. Dierks Lumber & Coal, Company was decided in favor of Lefevers. The case was decided on the merits and the jury could not reach a verdict. The jury was unable to reach a decision on the issue of liability.
ROUGE Scores 71: {'rouge1': Score(precision=0.125, recall=0.8333333333333334, fmeasure=0.21739130434782608), 'rouge2': Score(precision=0.10256410256410256, recall=0.8, fmeasure=0.18181818181818182), 'rougeL': Score(precision=0.125, recall=0.8333333333333334, fmeasure=0.21739130434782608)}

Original Text 46: State National Bank v. Fisher
Name Abbreviation 46: State National Bank v. Fisher
Generated Summary 46: State National Bank v. Fisher is a case involving a state-owned bank. State National Bank is a bank based in New York City, New York. The case was brought by the State of New York against Fisher Bank.
ROUGE Scores 46: {'rouge1': Score(precision=0.1282051282051282, recall=1.0, fmeasure=0.22727272727272727), 'rouge2': Score(precision=0.10526315789473684, recall=1.0, fmeasure=0.1904761904761905), 'rougeL': Score(precision=0.1282051282051282, recall=1.0, fmeasure=0.22727272727272727)}

Document Classification:

Classified legal opinions using a snippet of the casebody.head_matter. The model achieved

perfect classification accuracy, effectively categorizing legal opinions based on content.

```
Predicted Label True Label
1124 majority majority
813 majority majority
1175 majority majority
534 majority majority
514 majority majority
... ...
1213 majority majority
1040 majority majority
1393 majority majority
1281 majority majority
277 majority majority

[317 rows x 3 columns]

Accuracy: 1.0

Classification Report:
      precision    recall  f1-score   support

majority      1.00      1.00      1.00        317

accuracy      1.00      1.00      1.00        317
macro avg      1.00      1.00      1.00        317
weighted avg   1.00      1.00      1.00        317
```

CONCLUSION

This project demonstrates the potential of NLP techniques in automating the classification and summarization of legal documents. By applying these models to the SCOTUS dataset, key information such as decision dates were accurately extracted, and legal documents were effectively categorized and summarized. The results indicate that NLP can significantly improve the efficiency and accuracy of legal document analysis, making it an invaluable tool for legal professionals.

RECOMMENDATIONS

Named Entity Recognition (NER) for Decision Date Extraction:

- **Recommendations:** Regularly monitor the model to ensure continued accuracy. Consider expanding the training data and exploring advanced models like BERT for potential improvements.
- **Next Steps:** Periodically re-evaluate the model with new data and document performance metrics.

Text Summarization with BART:

- **Recommendations:** Fine-tune the BART model on legal texts and consider hybrid summarization methods to improve precision and F1 scores.
- **Next Steps:** Augment the dataset with more diverse examples and incorporate feedback from legal professionals.

Document Classification for Opinion Type:

- **Recommendations:** Validate the model on larger and more diverse datasets, and review edge cases to ensure robustness.
- **Next Steps:** Test scalability with larger datasets, implement continual learning mechanisms, and integrate the model into legal workflows.

FUTURE SCOPE OF WORK

- **Utilize RAG:** to add more cases to this dataset and refresh the results.
- **Utilize Citations to and from:** With guidance from some professionals with legal knowledge we can connect the cases which were related via citations to understand the citation network and identify most important cases with historical decisions within this data set.
- **Expand Model Generalization:** Future work should focus on expanding the model's ability to generalize across different types of legal documents and jurisdictions.
- **Enhance Data Annotation:** Improving the quality and quantity of annotated data will help in fine-tuning models and improving their accuracy.
- **Improve Summarization Techniques:** Continue exploring advanced summarization techniques to achieve more concise and relevant summaries, potentially using a combination of extractive and abstractive methods.